Debajit Biswas

SENIOR SOFTWARE ENGINEER | BACKEND SYSTEMS | HEALTHCARE INTEROPERABILITY (FHIR)

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// PROFESSIONAL SUMMARY

Software Engineer with 6 years of experience building scalable, reliable applications across healthcare and retail domains. Strong background in backend systems, distributed architectures, and cloud-based solutions. Acts as a go-to engineer for key components, contributing to system design and technical decisions while driving end-to-end delivery in Agile environments. Leverages AI assisted workflows to improve development velocity while maintaining high-quality, production ready systems. Currently expanding expertise in healthcare workflows and interoperability standards.

// TECHNICAL SKILLS

Languages

C# | TypeScript | JavaScript | SQL | Python | Powershell

Frameworks

.NET | Entity Framework Core | Angular | Durable Function

Testing

XUnit | Unit Testing | Postman | Jasmine

AI

GitHub Copilot | Claude CLI | BMAD Framework | SPEC Driven Developmet | MCP Server

Data

MS SQL Server | MariaDB | Azure Data Factory | Elastic Search | Power BI

Operations

App Dynamics | Azure DevOps | Azure Monitor | App Insights

Infrastructure

Azure | Terraform | Docker | CI/CD

Healthcare

Clinical Workflow | FHIR | Regulatory Reporting | OMOP | QDM

// EXPERIENCE

Altera Digital Health (formerly Allscripts)

Senior Software Engineer — Apr 2025 – Present

Enterprise Azure · Healthcare systems · FHIR platform architecture

- **Healthcare data engineering** — Led an independent POC within ACDL to transform clinical data into standardized models (OMOP + Extension →Proprietary model → QDM) for eCQM workflows, and identified architectural limitations that influenced a shift toward FHIR-based CQL approaches
- **Healthcare domain expertise** — Developed end-to-end understanding of regulatory workflows by working on AAP, a legacy suite for clinical data processing, eCQM measure computation, and QRDA reporting, gaining deep expertise in healthcare data pipelines and compliance-driven systems
- **FHIR-based modernization of TouchWorks** — Contributing to EHR modernization by mapping TouchWorks' legacy data models to FHIR standards, designing and implementing data ingestion pipelines using Azure Service Bus, Queue triggers, and Azure Data Factory to support multiple processing modes including historical loads, incremental updates, and reprocessing workflows
- **System optimization** — Co-architected complex stateful data processing into stateless, file-independent workflows, simplifying orchestration logic, significantly improving processing performance, reducing compute costs, and enabling more scalable and reliable data ingestion
- **FHIR platform & interoperability** — Set up and integrated a Medplum-based FHIR server to ingest and expose clinical data, enabling outbound data flows to downstream systems for eCQM computation using CQL, facilitating standardized data exchange across business units
- **Engineering productivity & reliability** — Built an internal GitHub repository migration tool using AI-assisted development with full code review, and introduced guardrail-based development practices to standardize workflows, reduce manual effort, and enable safer changes across repositories

Azure | FHIR | CQL | CI/CD | .NET | Angular | ADF | Terraform | Medplum | Docker

LTIMindtree

Senior Software Engineer Jul 2022 – Mar 2025

Retail · Enterprise systems · DevOps

- **System ownership & delivery** —Led development and ongoing maintenance of multiple Azure-hosted enterprise applications across data processing, reporting, and monitoring domains, ensuring reliable end-to-end operation
- **Full-stack engineering** — Designed and implemented RESTful APIs using .NET and Entity Framework Core, along with responsive Angular-based interfaces, enabling seamless integration across frontend, backend, and external systems with adequate documentation
- **Architecture & scalability** — Applied CQRS with MediatR and implemented asynchronous processing using Azure Queue messaging, improving performance, scalability, and resilience of data-intensive workflows
- **Security & access control** — Implemented authentication and authorization using Microsoft Entra ID and JWT, ensuring secure access and compliance with enterprise standards
- **Cloud & infrastructure** — Contributed to Azure service selection, resource provisioning, and network security setup, ensuring secure and compliant deployment environments
- **Code quality & testing** — Improved maintainability through unit testing (xUnit), structured object mapping (AutoMapper), and participation in code reviews
- **Cross-team coordination** — Acted as the primary owner for application maintenance, collaborating with multiple teams (product, cloud, data, and support) to triage issues, align dependencies, and ensure timely resolution of production and delivery needs

.NET | MS SQL Server | Elasticsearch | Azure Services | Even Driven Architecture | Rest API | Angular | React | IaC | GraphQL | ADF

Mindtree

Software Engineer Jun 2020 – Jun 2022

DevOps · Client delivery · Enterprise systems

- **Production delivery** — Contributed to a promotion planning and management platform for Unilever, supporting real-time business workflows and client requirements end-to-end
- **Feature development & maintenance** — Delivered enhancements, bug fixes, and performance improvements across application layers, ensuring consistent system reliability
- **Proactive improvements** — Identified and resolved recurring issues and inefficiencies, reducing manual effort and improving overall team productivity

.NET | Angular | SQL Server | Elasticsearch | Azure | ASPX | Rest API

Mindtree

Trainee Jan 2020 – May 2020

- **Foundation building** — Built foundational skills in .NET, Angular and MS SQL Server through dedicated self-learning, setting a strong base for technical growth
- **Team collaboration** — Developed a collaborative approach working closely with team members to deliver a full end-to-end demo e-commerce application

.NET | Angular | MS SQL Server

// INDEPENDENT WORK

Samit Roy Welfare Foundation Application NGO

Lead Developer Sep 2024 – Present

Non-profit · Full-stack · VPS hosting

- **Technical ownership** — Own the end-to-end development and ongoing operation of a live production platform supporting donations, events, and content workflows including responsive user-facing interfaces
- **System architecture** — Designed system architecture and data model using a code-first approach, enabling scalable and maintainable evolution of both backend services and UI integration
- **Infrastructure & reliability** — Responsible and accountable for hosting, deployment, and application security, ensuring reliable availability and secure handling of financial transactions
- **Product collaboration** — Work directly with stakeholders to drive feature evolution and make pragmatic technical decisions aligned with real needs

.NET | Angular | MariaDb | VPS | Linux

// CERTIFICATIONS & RECOGNITION

AZ-900 — Azure Fundamentals, Microsoft — December 2022

Highlighted — By manager for extraordinary work in eCQM POC

Recognition — Received multiple internal awards and client appreciation for consistent delivery and high-impact contributions

// EDUCATION

Meghnad Saha Institute of Technology — Kolkata, India

Bachelor of Technology in Electrical Engineering — DGPA 8.09 · 2015 – 2019

Dakshin Chatra High School — North 24 Parganas, India

Higher Secondary with PCM — 77.00% · 2014